Jack Norris

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EDUCATION

College of Computing, Georgia Institute of Technology — **Atlanta, GA**
***Degree**:* M.S. Computational Science & Engineering — *Expected December 2027*

College of Engineering, University of Arkansas (UARK) — **Fayetteville, AR**
***Degree**:* B.S. Computer Science (Summa Cum Laude), Minors in Math & Finance — *Graduated May 2024*

TECHNICAL PROFICIENCIES

- **Languages & Frameworks:** Python, C++, TypeScript, Java, SQL, Bash, P4, React, Spring Boot, PyTorch
- **Systems & Cloud**: Docker, Kubernetes, AWS, Linux, Firebase

TECHNICAL & PROFESSIONAL EXPERIENCE

University of Asunción AI/ML Drug Discovery Lab — **Asunción, Paraguay**
Research Fellow under Dr. Diego Galeano — *May 2025 – July 2026*

- Trained, tuned, & benchmarked 5 AI/ML models across 6 molecular representations for antibiotic discovery.
- Published first-author conference paper at IBERAMIA ’26 while serving as a Peace Corps Volunteer.

Compound Foundry — **Fayetteville, AR**
Software Engineering Intern — *August 2023 – February 2024*

- Built a Next.js app automating print-on-demand fulfillment, handling webhooks & third-party API integrations.
- Deployed canvas rendering for multi-format artwork; built a server-side print queue to manage fulfillment.

UARK Computer Networks Lab — **Fayetteville, AR**
Undergraduate Researcher under Dr. Kevin Jin — *May 2022 – May 2024*

- Wrote data plane logic in P4 to parse PMU network traffic & detect packet loss; built a custom Python packet generator/receiver to validate packet throughput & recovery performance.
- Published first-author conference paper at IEEE SmartGridComm 2025.

Amazon — **Seattle, WA & Los Angeles, CA**
Software Engineering Intern — *Summer 2021 & Summer 2022*

- Engineered full-stack configuration management service using Scala & React, decoupling manual deployment pipelines & cutting merchant form configuration turnaround time by 30%.
- Implemented real-time session synchronization for Amazon Music using Swift & GroupActivities, achieving sub-second sync latency on 4+ devices.

J.B. Hunt Transport Services — **Fayetteville, AR**
Software Engineering Intern — *Fall & Spring 2020 – 2022*

- Architected a driver app feedback pipeline via React/Firebase, aggregating real-time ratings for 15k+ users.
- Co-engineered a full-stack internal interview feedback platform using Angular & Spring Boot.

PROFESSIONAL DEVELOPMENT

Peace Corps Paraguay Volunteer: Built community outdoor gym, computer lab, & public library in rural Paraguay
***Association for Computing Machinery President*:** Organized hackathon with ~200 participants & $13k funding
***McMillon Innovation Studio Project Lead*:** Led technical prototyping for UARK startups, secured $20k funding
NWA Children’s Hospital Volunteer: 1.5 years entertaining patients, cleaning toys, & escorting families to pre-op

SELECTED PROJECTS

Lightweight Automatic Differentiation Engine

- Built a lightweight automatic differentiation engine in Python, implementing computational graphs & backpropagation from scratch; trained a custom neural network achieving 87.5% accuracy on MNIST dataset.

Qwirkle Online

- Developed & deployed a full-stack multiplayer online boardgame; containerized a Node.js WebSocket service with Docker to support low-latency, bidirectional event streaming.

Vector Network Analyzer (VNA) Interface App

- Architected and built a custom Kotlin Android app that interfaces via Bluetooth with a VNA, supporting advanced nanotechnology research focused on detecting pathogens in communal spaces.

8-Bit Breadboard Computer

- Constructing an 8-bit breadboard CPU using discrete logic gates; completed the variable clock circuit, ALU, registers, & addressable RAM.